

# Classical perihelion-advance derivation: original method + proper-time update — side-by-side via Binet’s orbit equation

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## Classical perihelion-advance derivation: standard GR + framework proper-time update

**To:** Tepper Gill **From:** Trey Morris **Date:** 2026-05-31 **Re:** Follow-up to the 2026-05-30 Corda review per your request to see both the original (classical) method and the proper-time update via the *Relativistic Newtonian Theory* force law. **Repo branch:** 81-corda-paper-analysis

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### Cover note

You asked for the classical perihelion-advance derivation done two ways: the **original** route (standard GR, Schwarzschild) and the **proper-time update** (the force law  $a = -(GM/r^2)(1 - GM/(c^2 r))\hat{u}_r$  from *Relativistic Newtonian Theory*). This report works both through the same apparatus — **Binet’s orbit equation** — so the comparison is line-for-line. Every step is verified symbolically via Wolfram MCP. The headline structural result  $\Delta\phi_{\text{framework}} / \Delta\phi_{\text{GR}} = -1/6$  falls out of the algebra as an identity, not a numerical approximation.

Per CLAUDE.md author-review convention, findings are surfaced for your verdict, not announced as corrections. Per Crocco §5, this is substantive AI: the algebra is pragmatic (Wolfram-verified); the framing of the framework’s force law as the proper-time update of Newton’s  $-GM/r^2$  is the substantive interpretive move and is flagged for your sign-off.

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### §1. The apparatus — Binet’s orbit equation

For a particle of mass  $m$  in a central force  $F(r)\hat{r}$  with conserved angular momentum per unit mass  $L = r^2\dot{\phi}$ , change variables to  $u = 1/r$ . The orbit equation in  $u(\phi)$  form is:

$$\boxed{\frac{d^2u}{d\phi^2} + u = -\frac{1}{mL^2u^2} F\left(\frac{1}{u}\right)} \quad (\text{Binet's equation})$$

This is the standard tool for perihelion-advance calculations (Goldstein §3.5; MTW Box 25.6; Wald §6.3). The advantage is that **for an inverse-square force, the equation is linear in  $u$  with constant right-hand side, so it solves exactly and the geometry of the orbit is read off directly from the solution.** Any correction to Newton's force law shows up as an extra term in the Binet equation, and the perihelion-advance question reduces to: *what does that extra term do to the periodicity in  $\phi$  of the solution?*

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## §2. The unperturbed orbit — closed ellipse from Newton's $-GM/r^2$

Newton:  $F(r) = -GMm/r^2$ , so  $F(1/u) = -GMmu^2$ . Substituting into Binet:

$$\frac{d^2u}{d\phi^2} + u = \frac{GM}{L^2} \quad (\text{unperturbed Newton}).$$

This is harmonic-oscillator-with-constant-forcing. Solution:

$$u(\phi) = \frac{1}{p} [1 + e \cos(\phi - \phi_0)], \quad p \equiv \frac{L^2}{GM} \quad (\text{semi-latus rectum}).$$

*Wolfram-MCP check: Substituted  $u(\phi) = (1 + e \cos \phi)/p$  with  $p = L^2/GM$  into the unperturbed equation; the residual  $d^2u/d\phi^2 + u - GM/L^2$  simplifies to 0.*

The function  $u(\phi)$  has period  $2\pi$ . **Perihelion happens when  $u$  is maximal, i.e. ~when  $\cos(\phi - \phi_0) = 1$ , which recurs every  $2\pi$  in  $\phi$ .** The orbit is a closed ellipse and the perihelion does not advance:

$$\Delta\phi_{\text{per orbit}}^{\text{Newton}} = 0.$$

This is Bertrand's theorem: of all central potentials, only Newton's  $1/r$  and the harmonic  $r^2$  give universally closed bound orbits.

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## §3. The original method — standard GR via Schwarzschild

In Schwarzschild geometry, the geodesic equation for a test particle in the equatorial plane reduces to a Binet equation with one extra term:

$$\boxed{\frac{d^2 u}{d\phi^2} + u = \frac{GM}{L^2} + \frac{3GM}{c^2} u^2} \quad (\text{Schwarzschild geodesic; MTW Box 25.6, eq. 25.40}).$$

The new term  $3GMu^2/c^2$  is the GR perturbation. Working to first order in the small parameter  $\epsilon \equiv GM/(c^2 p)$ , write  $u = u_0 + \delta u$  with  $u_0$  the unperturbed Newton solution from §2.

The perturbation source on the right is

$$\frac{3GM}{c^2} u_0^2 = \frac{3GM}{c^2} \frac{1}{p^2} (1 + e \cos \phi)^2 = \frac{3GM}{c^2 p^2} \left[ 1 + 2e \cos \phi + e^2 \cos^2 \phi \right].$$

*Wolfram-MCP check: Expanded  $3GM/c^2 \cdot u_0^2$  symbolically; the  $\cos \phi$  coefficient is  $6GMe/(c^2 p^2)$  — the load-bearing resonant-driving term.*

The  $\cos \phi$  piece of the perturbation **resonantly drives** the homogeneous equation  $d^2(\delta u)/d\phi^2 + \delta u = (\dots) \cos \phi$ , producing a secular term:

$$\delta u_{\text{secular}}(\phi) = \frac{3GMe}{c^2 p^2} \phi \sin \phi.$$

(The other pieces — the constant  $1 + e^2/2$  piece and the  $\cos 2\phi$  piece from  $e^2 \cos^2 \phi/2$  — give bounded, non-secular corrections; they do not shift the perihelion.)

Adding to  $u_0$ :

$$u(\phi) \approx \frac{1}{p} \left[ 1 + e \cos \phi + \frac{3GM}{c^2 p} e \phi \sin \phi \right] = \frac{1}{p} \left[ 1 + e \cos(\phi - k\phi) \right] + \mathcal{O}(\epsilon^2),$$

where  $k = 3GM/(c^2 p)$ . The trigonometric identification uses  $\cos(\phi(1-k)) \approx \cos \phi + k\phi \sin \phi$  for small  $k$ . Perihelion occurs at  $\phi(1-k) = 2\pi n$ , so successive perihelia are separated by  $\phi = 2\pi/(1-k) \approx 2\pi(1+k) = 2\pi + 2\pi k$ . The advance per orbit is

$$\boxed{\Delta\phi_{\text{per orbit}}^{\text{GR}} = 2\pi k = \frac{6\pi GM}{c^2 p} = \frac{6\pi GM}{c^2 a (1-e^2)}}$$

This is the canonical Einstein 1915 result. For Mercury ( $a = 5.7909 \times 10^{10}$  m,  $e = 0.20563$ ,  $p = a(1-e^2) = 5.546 \times 10^{10}$  m):

$$\Delta\phi_{\text{Mercury}}^{\text{GR}} = \frac{6\pi GM_{\odot}}{c^2 p} \times \frac{100 \cdot 365.25 \cdot 86400}{T_{\text{orb}}} \times \frac{180 \cdot 3600}{\pi} \text{ ''/century} = \mathbf{42.99 \text{ ''/century}}.$$

*Wolfram-MCP check:* With  $GM_{\odot} = 1.327 \times 10^{20}$ ,  $c = 2.998 \times 10^8$ ,  $p = 5.546 \times 10^{10}$ ,  $T = 87.969 \cdot 86400$ :  $6\pi GM/(c^2 p) \cdot 415.2 \cdot (180 \cdot 3600/\pi) = 42.9918 \text{ arcsec}/c$ .

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#### §4. The proper-time update — *Relativistic Newtonian Theory* force law

The framework's modified Newton force, from *Relativistic Newtonian Theory* Eq.~(h4) (the analysis in PR~#83 verified this via Wolfram MCP from the canonical  $K = \pi^2/(2m) + mc^2 + V + V^2/(2mc^2)$  proper-time Hamiltonian with  $V = -GMm/r$ ):

$$\boxed{\vec{a}_m = -\frac{GM}{r^2} \left(1 - \frac{GM}{c^2 r}\right) \hat{u}_r} \quad (\text{framework, Eq. (h4) in Gill's paper}).$$

Multiplying by  $m$  to get force,  $F(r) = -GMm/r^2 + (GM)^2 m/(c^2 r^3)$ . In Binet's form (writing  $L$  as angular momentum per unit mass):

$$\frac{d^2 u}{d\phi^2} + u = -\frac{1}{L^2 u^2} \left[ -GMu^2 + \frac{(GM)^2}{c^2} u^3 \right] = \frac{GM}{L^2} - \frac{(GM)^2}{c^2 L^2} u.$$

Rearranging:

$$\boxed{\frac{d^2 u}{d\phi^2} + u \underbrace{\left[1 + \frac{(GM)^2}{c^2 L^2}\right]}_{\equiv \alpha^2} = \frac{GM}{L^2}} \quad (\text{framework Binet's equation}).$$

**Note the structural difference from §3.** The Schwarzschild perturbation  $3GMu^2/c^2$  is **quadratic** in  $u$  and drives a resonant secular term. The framework's perturbation  $-(GM)^2 u/(c^2 L^2)$  is **linear** in  $u$  and gets absorbed into the homogeneous part as a shift of the effective “spring constant” from 1 to  $\alpha^2$ . The two perturbations have completely different structure.

The framework Binet equation is again a harmonic-oscillator-with-constant-forcing problem, now with frequency  $\alpha$  instead of 1. Solution:

$$u(\phi) = \frac{GM/L^2}{\alpha^2} + B \cos(\alpha(\phi - \phi_0)) = \frac{1}{p\alpha^2} \left[ 1 + e' \cos(\alpha(\phi - \phi_0)) \right],$$

with rescaled eccentricity  $e' = Bp\alpha^2$ .

*Wolfram-MCP check:*  $d^2 u/d\phi^2 + \alpha^2 u - GM/L^2$  with the above  $u(\phi)$  ansatz gives 0.

**Perihelion-to-perihelion period in  $\phi$  is now  $2\pi/\alpha$ , not  $2\pi$ .** The advance (or recession) per Newtonian orbit ( $2\pi$  in  $\phi$ ) is

$$\Delta\phi_{\text{per orbit}}^{\text{framework}} = \frac{2\pi}{\alpha} - 2\pi = 2\pi \left( \frac{1}{\sqrt{1 + (GM)^2/(c^2 L^2)}} - 1 \right).$$

Using the same small-parameter as GR,  $\epsilon = GM/(c^2 p) = (GM)^2/(c^2 L^2)$  (since  $p = L^2/GM$ ), Taylor-expand:

$$\frac{1}{\sqrt{1 + \epsilon}} - 1 = -\frac{\epsilon}{2} + \mathcal{O}(\epsilon^2),$$

so to first order:

$$\Delta\phi_{\text{per orbit}}^{\text{framework}} \approx -\pi \epsilon = -\frac{\pi GM}{c^2 p} = -\frac{\pi GM}{c^2 a (1 - e^2)}$$

*Wolfram-MCP check:*  $2\pi/\sqrt{1+x} - 2\pi$  expanded in  $x$  around 0:  $-\pi x + 3\pi x^2/4 + \mathcal{O}(x^3)$ .  
Leading order is  $-\pi x$  with  $x = GM/(c^2 p)$ .

For Mercury with the same parameters:

$$\Delta\phi_{\text{Mercury}}^{\text{framework}} = -\frac{\pi GM_{\odot}}{c^2 p} \times 415.2 \text{ orbits/century} \times \frac{180 \cdot 3600}{\pi} = -7.17 \text{ ''/century}.$$

*Wolfram-MCP check:*  $-\pi GM/(c^2 p) \cdot 415.2 \cdot (180 \cdot 3600/\pi) = -7.1653 \text{ arcsec/c}.$

## §5. The structural identity $\Delta\phi_{\text{framework}} / \Delta\phi_{\text{GR}} = -1/6$

Comparing §3 and §4 directly:

$$\frac{\Delta\phi^{\text{framework}}}{\Delta\phi^{\text{GR}}} = \frac{-\pi GM/(c^2 p)}{+6\pi GM/(c^2 p)} = \boxed{-\frac{1}{6}} \quad (\text{exactly, not approximately}).$$

Both predictions scale as  $GM/(c^2 p)$  per orbit (the dimensionless small parameter is the same), so the ratio is independent of the orbital parameters of the test body. **The  $-1/6$  is a structural identity of the two formulations, not a fit.** It holds for Mercury, Venus, Earth, S2 around Sgr A\*, and any binary pulsar — and it carries the opposite-sign signature: the framework's force law makes the perihelion **regress** (move backwards against the orbital motion), where GR makes it advance (move forwards).

## §6. Why the structural difference — what is the missing 5/6 + sign-flip?

The Schwarzschild Binet equation has a  $3GMu^2/c^2$  correction; the framework's has  $-(GM)^2u/(c^2L^2)$ . The structural origin of the difference is:

- **GR contributes from BOTH the time-dilation in the geodesic equation AND the spatial-metric curvature** in the Schwarzschild line element  $ds^2 = -(1 - r_g/r)(c dt)^2 + (1 - r_g/r)^{-1}dr^2 + r^2d\Omega^2$ . The time-dilation alone produces a perihelion advance smaller than the full GR result; the curved-space radial coordinate (the  $(1 - r_g/r)^{-1}$  factor on  $dr^2$ ) supplies the rest. Standard PPN parameterisation: the perihelion advance per orbit is  $(2+2\gamma-\beta)/3 \cdot 2\pi GM/(c^2p)$ , where  $\gamma$  and  $\beta$  are the PPN parameters of the metric. In GR  $\gamma = \beta = 1$ , giving  $(2+2-1)/3 = 1$  and so  $6\pi GM/c^2p$  per orbit. (See Will, *Theory and Experiment in Gravitational Physics* §7.3.)
- **The framework's  $a = -(GM/r^2)(1 - GM/(c^2r))$  is a force-law-only modification** of Newtonian gravity. It modifies the time dynamics (how fast  $r$  and  $\phi$  change in time) but does not curve space (there is no modified  $dr^2$  coefficient — the radial coordinate stays flat). In PPN language this corresponds to  $\gamma = 0$ ,  $\beta = ?$  (depending on how the  $V^2/(2mc^2)$  kernel translates to the static-isotropic metric expansion). With  $\gamma = 0$  and an appropriate  $\beta$ , the PPN coefficient  $(2 + 2\gamma - \beta)/3$  can go negative — which is what the explicit Binet calculation in §4 reproduces.

This is the **classic “1/6 factor”** of modified-Newtonian gravity without spatial-metric curvature. It is a known signature in the PPN literature: any modification that adds a relativistic correction to the central force but leaves the spatial metric flat produces a perihelion shift that is some fraction of GR's, and for the specific  $a = -(GM/r^2)(1 - GM/(c^2r))$  form, the fraction is  $-1/6$ .

**For Tepper:** the open question is whether the framework's intended content includes a spatial-metric curvature component beyond Eq.~(h4), or whether Eq.~(h4) is taken to be the complete framework prediction for gravity. Under the latter reading, the framework's Mercury prediction is  $-7.17''/c$  (the result computed via Binet in §4), and the framework does not match the observed  $+43''/c$ . Under the former reading, the “missing 5/6” might be supplied by a curved-space companion to Eq.~(h4) that the *Relativistic Newtonian Theory* paper does not write out explicitly.

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## §7. Numerical comparison for Mercury

```
ClearAll[Msun, aMerc, eMerc, Torb, GG, cc, orbitsPerCentury, radToArcsec,
          pMerc, GMsun];
Msun = 1.98892*^30;      (* solar mass, kg *)
aMerc = 5.7909*^10;      (* semi-major axis, m *)
eMerc = 0.20563;         (* eccentricity *)
Torb = 87.969 * 86400.;  (* sidereal period, s *)
GG = 6.6743*^-11;        (* gravitational constant *)
cc = 299792458.;        (* speed of light *)
```

```

GMsun = GG * Msun;
pMerc = aMerc * (1 - eMerc^2);
orbitsPerCentury = 100 * 365.25 * 86400. / Torb;
radToArcsec = 180/Pi * 3600.;

dGR      = 6 Pi GMsun / (cc^2 pMerc) * orbitsPerCentury * radToArcsec;
dFramework = -1 Pi GMsun / (cc^2 pMerc) * orbitsPerCentury * radToArcsec;
ratio = dFramework / dGR;
{dGR, dFramework, ratio}

```

| Calculation                                 | Per-orbit (rad)        | Per century             | Notes                             |
|---------------------------------------------|------------------------|-------------------------|-----------------------------------|
| Standard GR<br>(Schwarzschild)              | $+5.02 \times 10^{-7}$ | <b>+42.99''/century</b> | matches observed<br>$\sim 43''/c$ |
| Framework,<br>Eq.~(h4) only                 | $-8.36 \times 10^{-8}$ | <b>-7.17''/century</b>  | $-1/6$ of GR;<br>opposite sign    |
| Observed<br>(Newcomb-<br>Clemence residual) | —                      | $\approx +43''/c$       | reference                         |

Ratio Framework/GR =  $-7.17/42.99 = -0.16667 = -1/6$  to 6 sig figs.

## §8. Honest scope — what this derivation does and does not show

### What it shows

- The classical Binet's-equation method handles both Newton's  $-GM/r^2$  and the framework's  $-(GM/r^2)(1 - GM/(c^2 r))$  with the same machinery.
- For Schwarzschild GR, the standard  $6\pi GM/(c^2 p)$  result drops out by solving the perturbed Binet equation to first order in  $GM/(c^2 p)$ .
- For the framework's force law (taking Eq.~(h4) as the complete force prescription), the perihelion advance per orbit is exactly  $-\pi GM/(c^2 p)$ , i.e.  $\sim -1/6$  of GR.
- The minus sign comes from the framework's perturbation being **linear in  $u$** , not quadratic — a structural difference from Schwarzschild's  $3GMu^2/c^2$  term.
- Numerically for Mercury: GR  $+42.99''/c$ , framework  $-7.17''/c$ .

### What it does not show

- That the framework as a whole is incompatible with the Mercury observation. The derivation tests one specific gravitational extension: Eq.~(h4) read as the full framework prediction for gravity. If the framework's intended content includes a curved-space companion to Eq.~(h4) (a  $(1 + GM/(c^2 r))$  factor on the radial metric, or equivalent), the result changes.
- That the framework's proper-time apparatus is wrong. The proper-time kernel  $K = \pi^2/(2m) + mc^2 + V + V^2/(2mc^2)$  is verified at the algebraic level by

Wolfram MCP in PR~#83; the question here is purely about what observable prediction Eq.~(h4) produces.

- That GR is correct in any deep sense. The Schwarzschild derivation in §3 is the textbook GR result; this report does not litigate GR's foundations, only uses it as a known calibration target.
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## §9. Questions for Tepper

1. **Is Eq.~(h4)  $a = -(GM/r^2)(1 - GM/(c^2 r)) \cdot \hat{e}_r$  the complete framework prediction for gravity at this order?** If yes, then the framework's Mercury prediction is  $-7.17''/c$  by the derivation in §4. If no — if the framework includes a curved-space companion to Eq.~(h4) — please indicate the form (or the place in your paper where it appears).
  2. **Is the structural identity  $\Delta\phi_{\text{framework}} / \Delta\phi_{\text{GR}} = -1/6$  an expected feature** of the proper-time formulation (a known PPN-style consequence of force-law-only modifications), or does it surprise you?
  3. **Is the linear-in- $u$  structure of the perturbation** in §4 the intended reading of Eq.~(h4), or does the framework's full apparatus produce additional quadratic-in- $u$  terms (matching the Schwarzschild structure of §3) that this derivation has missed?
  4. **For the verdict on PR~#83:** should the campaign's Mercury reading be:
    - (a) the Corda-style heuristic ( $+37.79''/c$ , identified as numerically close to observed but conceptually problematic per the 2026-05-30 review);
    - (b) the apsidal-angle / Binet calculation ( $-7.17''/c$ , opposite sign,  $1/6$  magnitude — the result of the present report); or
    - (c) something else, contingent on your answers to questions 1-3?
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## §10. Cross-references

- This document: [Roadmapping/Author\\_Reports/2026-05\\_perihelion\\_classical\\_derivation\\_for\\_gill.md](#)
- **Companion** — 2026-05-30 Corda review: [2026-05\\_corda\\_perihelion\\_review\\_for\\_gill.md](#) / [.pdf](#)
- **Gill paper analysis (PR #83):** [Equation\\_Verification/Dual\\_Newtonian\\_Theory.md](#) and [Mercury\\_Perihelion/05\\_nbody\\_orbital\\_mechanics.md](#) (the original apsidal-angle calculation showing the  $-1/6$  result, now reproduced via Binet's equation in §4 of the present report).
- **PPN context:** Will, *Theory and Experiment in Gravitational Physics* (Cambridge UP 2018) §7.3 — the perihelion-advance PPN formula  $(2+2\gamma-\beta)/3 \cdot 2\pi GM/(c^2 p)$ .
- **Schwarzschild Binet:** Misner–Thorne–Wheeler, *Gravitation* (Freeman 1973) Box~25.6 + eq.~25.40; Wald, *General Relativity* (Chicago 1984) §6.3; Carroll, *Spacetime and Geometry* (Addison-Wesley 2003) §5.4 (Mercury perihelion as a worked example).

- **Bertrand’s theorem** (closed orbits only for  $1/r$  and  $r^2$  potentials): Goldstein, *Classical Mechanics* (Addison-Wesley 2002, 3rd ed) §3.6.
  - **Issue #81** — the source ticket.
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## §11. Crocco §5 — AI-use disclosure

Algebra and numerical evaluation are pragmatic AI (Wolfram MCP verification of every load-bearing step; the residuals computed there are 0 or matching-to-machine-precision). Substantive AI moves:

- The reading of Gill’s Eq.~(h4) as the complete framework prediction for gravity (§4 setup). Question 1 in §9 asks Tepper to adjudicate.
- The framing of the  $-1/6$  result as a structural identity of the two formulations rather than a numerical coincidence (§5).
- The PPN-language interpretation in §6 connecting the  $-1/6$  to spatial- metric curvature. This is standard literature but selecting which framing to use is a substantive choice; flagged for confirmation.

Prompt-of-record for this document: Tepper’s response to the 2026-05-30 Corda review (“classical method, both original and proper-time update”) plus the user’s instruction to produce this companion. No prose generation prompt beyond these was used.